

Current Understanding of Nephrotic Syndrome in Children



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KEYWORDS

• Children • Nephrotic syndrome • Minimal change • Glomerular disease • Edema • Anasarca

KEY POINTS

- Children with nephrotic syndrome have edema and massive proteinuria.
- Steroid therapy is the standard of care.
- Kidney biopsy is needed in very few patients.
- Prognosis is excellent in most patients.
- Relapses may continue into adult life.

INTRODUCTION

Nephrotic syndrome (NS) in children is characterized by the presence of significant proteinuria, hypoalbuminemia, and edema. About 90% of the cases between 1 and 10 years of age and 50% of the cases above 10 years of age are idiopathic (INS) in origin.¹ Most patients with INS respond to corticosteroid therapy; 80% to 90% of patients relapse and nearly half of them have frequent relapses (FR) and may become steroid dependent (SD). Additional immunosuppressive therapy is needed in those with steroid resistance or FR/SD clinical course. The overall prognosis in those who respond to steroids is excellent. About a third of children with INS diagnosed during childhood, particularly those with more severe form of the disease, may have relapses during adult life. Many country-specific practice guidelines have been published and most of them are very similar with clinically insignificant differences. This chapter is focused primarily on steroid-sensitive INS; secondary NS and steroid resistant NS are not discussed here.

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HISTORICAL ASPECTS

Edema, referred to as dropsy, from the Greek *hydrops* (water) has been documented in the medical literature from the times of Hippocrates about 2400 years ago. More than 500 years ago (1484), Cornelius Roelans of Belgium is credited with the first crude description in a child with NS: "The fifty-first disease of children is swelling of the whole body of the child.", very likely referring to nephrotic edema. Perhaps one of the best early descriptions of the NS was made 250 years later by the pediatrician Theodore Zwinger III of Basel in his *Paedoiatreia practica* of 1722. Despite this detailed clinical description of NS, Zwinger does not mention any associated urinary abnormalities. Almost 50 years later, Domenico Cotugno in his *De ischiade nervosa commentarius* of 1764, was the first to describe proteinuria upon boiling the urine of a 28-year-old man with dropsy and document albuminuria in NS and diabetic nephropathy. Fifty years later (1827), Richard Bright and others put together the triad of generalized edema, proteinuria, and kidney disease and thus are credited with describing NS in all its details. The term "nephrosis" was first mentioned by Friedrich von Muller of Marburg in 1905 (12). This term was popularized by Fritz Munk and for some time the term Munk's lipid nephrosis was used to describe these cases. About the same time, Henry Christian and Louis Leiter in the United States pointed out the similarities of primary and secondary forms of severe proteinuric disease with edema, and introduced the terms "syndrome of nephrosis," or "nephrotic syndrome."^{2,3}

EPIDEMIOLOGY

The overall incidence of NS in children is about 3/100,000 and ranges from 2% to 8% in various parts of the world. It is lowest in white children in Europe and North America and higher in South, Southeast and East Asian, Middle Eastern, and Northern African children. It is nearly twice as common in black as compared to white children. Its prevalence is about 16/100,000.⁴⁻⁶ In younger children, it is more common in males (approximately 2:1). Only about 3% of patients have affected siblings,⁷ and the incidence is 2 to 3 times higher in communities with a high rate of consanguinity.⁸

PATHOPHYSIOLOGY

Proteinuria

The underlying renal defect in NS is an increased glomerular basement membrane (GBM) permeability to albumin. Normally, very little (less than 0.1%) plasma albumin, which has a large molecular weight, crosses the glomerular filtration barrier and most of it is reabsorbed by the proximal tubular cells. Compared with a normal urine albumin excretion of usually less than 30 to 50 mg/24 h/1.73 m²,⁹ the albuminuria in NS may increase 100 to 200-fold. This massive albuminuria eventually leads to hypoalbuminemia. The increased catabolism of albumin in the renal tubular cells also contributes to hypoalbuminemia. Hypoalbuminemia causes a decrease in plasma oncotic pressure and edema (see below). The liver increases albumin synthesis but not enough to compensate for urine losses and increased catabolism.

The glomerular filtration barrier is made up of 3 layers (**Fig. 1**): (1) a thin fenestrated endothelial layer, (2) GBM, and (3) a visceral epithelial layer of cells known as podocytes. The podocytes have foot processes, which are very dynamic and functional cellular extensions that rest on the GBM. They are interconnected by the slit diaphragms to form the final component of the kidney permeability barrier.⁶ Podocyte injury is at the core of significant proteinuria in NS. In view of its importance and as shown in **Table 1**, there is a trend to classify NS according to the underlying etiology for podocyte injury (podocytopathy).¹⁰

Podocyte



□ PP = primary processes
 □ FP = foot processes

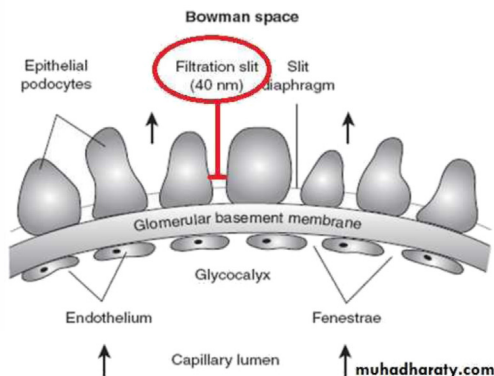


Fig. 1. The glomerular filtration barrier: (1) thin fenestrated endothelial layer, (2) glomerular basement membrane, and (3) a visceral epithelial layer of cells known as podocytes. (From Leeuwis JW, Nguyen TQ, Dendooven A, et al. Targeting podocyte-associated diseases. *Advanced Drug Delivery Reviews* 2010;62(14) pp 1325-1336.)

Previously it was believed that the INS is exclusively a T-cell dysfunction associated with a circulating factor or lymphokine. This hypothesis was based on the following clinical observations: (1) remission was commonly observed with measles infection whereby cell-mediated immunity was suppressed; (2) NS with minimal change is associated with Hodgkin's disease, which is a known T-cell disorder; and (3) patient response to corticosteroids and cyclophosphamide, which are inhibitors of T-cell function.¹¹ However, nearly 2 decades ago, it became obvious that B-lymphocytes also have a role in INS. The most compelling evidence is the sustained remission of NS following administration of rituximab, an anti-CD20 monoclonal antibody that depletes B cell population, and the occurrence of NS in some people with light chain gammopathy.¹²⁻¹⁴ A recent study reported the presence of circulating nephrin autoantibodies in some children and adults with minimal change NS (MCNS). These were present during active disease and decreased significantly or were absent during treatment response.¹⁵

Table 1
Genetic and secondary causes of podocyte injury

Intrinsic: Genetic Mutations	Extrinsic: Podocyte Stress
Nucleus: <i>WT1</i> , <i>PAX2</i> , and <i>LMX1B</i> mutations	Viral: infection or circulating viral protein
Cytoskeleton: <i>ACTN4</i> , <i>MYH9</i> , <i>INF2</i> , <i>ANLN</i> , <i>ARHGA24</i> mutations	Toxic: medication (pamidronate, interferon, toxin) (puromycin amino nucleoside, adriamycin)
SD complex: <i>NPHS1</i> , <i>NPHS2</i> , <i>CD2AP</i> , <i>PLCE1</i> , <i>TRPC6</i> mutations	Lymphokine or another host protein: IFN- α , IFN- β , FSGS permeability factor
GBM: <i>LAMB2</i> , <i>ITGB4</i> , <i>ITGA3</i> , <i>COL4A3/4/5</i> mutations	Mechanical: podocyte stretch (glomerulomegaly)
Mitochondria: <i>tRNA</i> , <i>COQ2</i> , <i>COQ6</i> , <i>COQ8</i> mutations	Acute ischemia with thrombotic microangiopathy
Metabolic, lysosomal: <i>GLA</i> , <i>OCRL</i> , <i>SCARB2</i> , <i>CFH</i> , <i>DGKE</i>	Immunologic: immune complex deposition or <i>in situ</i> formation (lupus, IgA nephropathy, membranous nephropathy)
	Metabolic: diabetes

Adapted and modified from Refs.^{10,22}

Edema

As shown in Fig. 2, the initial trigger for edema formation is the decrease in plasma oncotic pressure below a critical level of 12 to 15 mm Hg, causing an imbalance in Starling’s forces at the capillary end and resulting in net increase in plasma translocation into the interstitial spaces¶. Edema is sustained by compensatory mechanisms resulting from the decreased effective blood volume, which may be associated with (a) reduced glomerular filtration and increase in sodium and water reabsorption and (b) activation of homeostatic factors including the renin–angiotensin–aldosterone axis and antidiuretic hormone, and (c) a decrease in natriuretic peptide (ANP), all leading to salt and water retention in the distal renal tubules. This pathway for edema formation is known as the underfill hypothesis and applies to most children with INS. The overfill hypothesis implies that there is primary sodium and water retention with a rise in capillary hydrostatic pressure and hypervolemia, which causes intravascular fluid transudation into the interstitium. The underlying mechanisms are believed to be an activation of the epithelial sodium channels (ENaC) and Na-K ATPase in the distal renal tubules.^{16,17}

DEFINITIONS

Nephrotic range proteinuria in children is defined by urine protein excretion of ≥ 40 mg/m² BSA/h, ≥ 1 gm/m² BSA/d, or greater than 50 mg/kg/d. However, it is neither

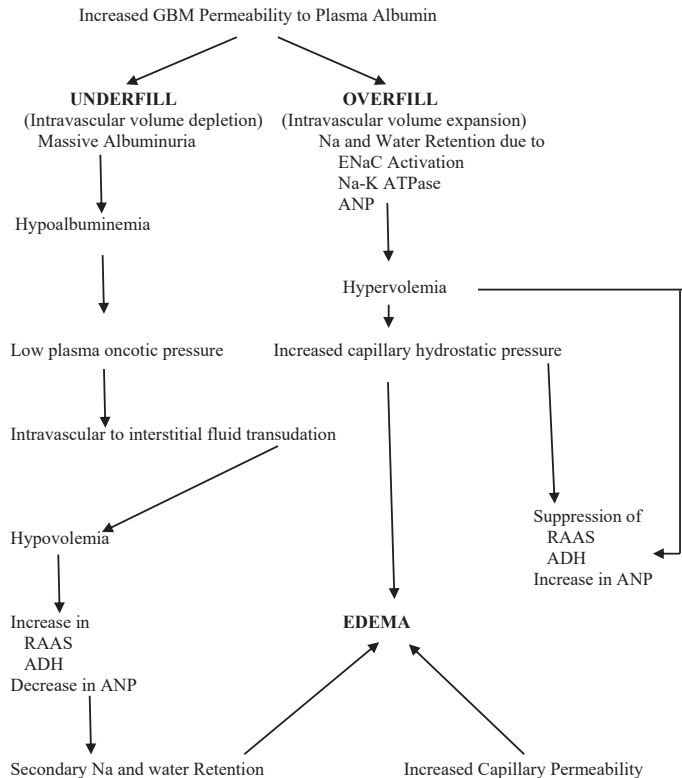


Fig. 2. Pathogenesis of edema and intravascular volume changes. ADH, antidiuretic hormone; ANP, atrial natriuretic peptide; ENaC, sodium epithelial channel; RAAS, Renin–angiotensin–aldosterone system.

practical nor necessary to have a timed urine collection for diagnosis. A random urine sample with $\geq 3^+$ protein (corresponding to ≥ 300 mg/dL) by dipstick examination or a protein/creatinine ratio of greater than 3 mg/mg is good enough for diagnosis. Serum albumin in patients with NS is usually less than 2.5 gm/dL.¹⁸ Some commonly used definitions in INS are given in [Table 2](#).

CLASSIFICATION

The 3 different classifications that have a strong bearing on clinical management of INS are based on the age of disease onset, underlying renal histopathology, and response to the initial steroid therapy.

Age of Disease Onset

According to the age of onset (not diagnosis), NS is classified as follows:

- Congenital: Birth to 3 months of age
- Infantile: Older than 3 months and up to 1 year of age
- Childhood: Older than 1 year of age
- Adult: Adult life

Most children with congenital or infantile NS have a genetic basis and poor outcome as compared with those with childhood NS.

Renal Histopathology

The International Study of Kidney Disease in Children (ISKDC) study, which included renal biopsy in 521 children with NS reported MCNS in about 76.4% cases, membranoproliferative glomerulonephritis in 7.5% and focal segmental glomerulosclerosis in 6.9% (FSGS). The remaining ~10% cases had proliferative glomerulonephritis, pure

Table 2
Commonly used definitions in NS

Remission	Urine dipstick negative or trace (<30 mg/dL) for protein on 3 consecutive days or urine protein/creatinine ratio ≤ 0.2 (mg/mg).
Partial remission	Decrease of proteinuria by $\geq 50\%$ or urine protein/creatinine between >0.2 and <2 g/g
Relapse	Urinary dipstick showing $\geq 3^+$ proteinuria (corresponding to ≥ 300 mg/dL on urinalysis), P/c >2 g/g, for 3 d?
Steroid-sensitive NS	Remission during daily prednisone of 60 mg/m ² BSA for 4 wk
Late responder	Remission at 6 wk of daily prednisone therapy for initial treatment
Steroid-resistant NS	
Early	No remission after first (initial) daily prednisone of 60 mg/m ² BSA for 4 wk
Late	No response to standard steroid therapy for a relapse for 4 wk after initial response
Infrequent relapsing NS	1 relapse within 6 mo or ≤ 3 relapses within 12 mo after completion of initial steroid therapy
Frequent relapsing NS	≥ 2 relapses within 6 mo or ≥ 4 relapses within 12 mo after completion of therapy in steroid-sensitive NS
Steroid-dependent NS	≥ 2 relapses during steroid therapy or within 2 wk of its completion

(UPCR) based on a first morning void or 24-h collection.

diffuse mesangial proliferation, focal and global glomerulosclerosis, membranous glomerulopathy, and chronic glomerulonephritis.¹ However, the major limitation with this study was that it included cases with infantile NS and the data were collected from 24 participating centers with poorly defined patient inclusion and exclusion criteria. In another study, the ISKDC reported that about 95% of children with MCNS respond to steroid therapy as compared with 28% of those with other glomerular abnormalities,¹⁸ indicating that the patients with INS who respond to the initial corticosteroid therapy have mostly MCNS but it also includes non-MCNS cases, notably FSGS. With increasing age, the risk for FSGS and membranous nephropathy increases. In adults, MCNS is prevalent in approximately 15% of patients with INS.

Response to Corticosteroids

More than 90% of children with INS respond to steroid therapy. A lack of response helps define the need for renal biopsy and subsequent clinical management, particularly immunosuppression. Response to initial steroid therapy is a better marker for long-term clinical outcome than the underlying renal histopathology, and steroid-sensitive patients have a much better prognosis compared with those with steroid resistance. Ethnic variations in response to corticosteroid therapy have been reported.^{19,20} However, the possibility of socioeconomic and other factors contributing to these variations cannot be ruled out.

CLINICAL PRESENTATION

Patients with INS present with massive proteinuria, hypoalbuminemia, and edema. None of the other conditions that cause generalized edema in children is associated with significant ($\geq 3^+$) proteinuria (Table 3). The onset of INS is often preceded by upper respiratory tract infection (URI). Microhematuria is present in about 20% of cases. Depending on the severity, patients may also have oliguria or anuria, pulmonary congestion, fever, or sepsis. Hypertension is rare at presentation and its presence, or the presence of gross hematuria or other clinical findings as shown in Table 3, are suggestive of secondary NS. Some patients, particularly teenagers and older, may present with nephrotic-range proteinuria and hypoalbuminemia with no edema. The reason for the absence of edema in such patients, generally attributed to a decreased serum oncotic pressure due to hypoalbuminemia, is not known.

DIAGNOSIS

In addition to a good history and physical examination, certain investigations are necessary to confirm diagnosis and rule out the possibility of secondary NS.

Laboratory Tests

The laboratory tests that are commonly used in the diagnosis of INS are shown in Table 4.

Kidney Biopsy

Kidney biopsy is indicated in patients who present outside the usual age group (1–12 years) or show atypical clinical and laboratory findings (see above) suggestive of a secondary NS. The commonest reason for a kidney biopsy, regardless of age, is the lack of response to steroid therapy initially or later (late nonresponsiveness).

Table 3
Clinical presentation

Causes of generalized edema in children	• Nephrotic syndrome • Congestive cardiac failure • Liver disease, • Protein malnutrition (kwashiorkor) • Kidney failure with severe fluid overload
Features indicative of secondary NS	
Clinical	• Gross hematuria • Hypertension • Skin rashes • Joint pain/tenderness • Worsening kidney function • AKI • Prolonged fever • Cardiac murmur • Hepatosplenomegaly • Lymphadenopathy • Syndromic features/dysmorphic • Age <1 or >12 y
Other	• High-risk patient: HIV • Abnormal serum C3 and or C4, positive ANA, positive hepatitis B surface antigen or hepatitis C antibody, high antistreptolysin O titer • Presence of renal cysts⁵⁶ • Associated with RTA⁵⁷

Abbreviations: ANA, antinuclear antibodies; RTA, renal tubular acidosis.

Renal Imaging

Renal ultrasonography is not routinely needed as the findings are nonspecific. If performed, however, it will likely reveal enlarged, edematous kidneys with normal echoic pattern. In a study by Gershen and colleagues, it was found that increased echogenicity may be associated with resistance to corticosteroid therapy.²¹ Renal ultrasound is indicated if the child has abnormal urinary sediment, gross hematuria, suspected renal vein thrombosis azotemia, and hypertension or if a kidney biopsy has been performed.

Genetic Testing

Routine genetic testing is not needed in the management of INS. It should be considered in children with steroid resistant nephrotic syndrome (SRNS) as close to 30% may have a monogenic cause for their disease and more than 70 genes have been identified to date.²² The probability of finding a causative mutation is inversely proportional to the age at onset of the disease. Thus, the earlier the disease is diagnosed, the greater the relevance for genetic screening. As expected, mutations in recessive genes are in early-onset disease and mutations in dominant genes are more often implicated in adult-onset disease. A positive family history or consanguinity in any age group is highly suggestive of monogenic SRNS and recessive mutations are likely to be found on screening.²³ The presence of extra-renal manifestations or syndromic features would also be an indication for mutational analysis. Indications and benefits of genetic testing are elaborated in [Table 5](#).

Table 4 Laboratory tests for idiopathic nephrotic syndrome		
Routine Labs	Test	Interpretation
Urine	Urinalysis	3–4+ protein on urine dipstick
	Urine microscopy	20% of children have >3 RBC/HPF
	Protein/creatinine ratio	>2 mg/mg
Blood	CBC	WBC increase with suspected infection; elevated hemoglobin and hematocrit and platelet levels with hemoconcentration and intravascular volume depletion.
	Electrolytes	Hyponatremia, sometimes with mild hyperkalemia may be present
	BUN	intravascular volume depletion/AKI
	Creatinine	intravascular volume depletion/AKI
	Plasma albumin	<2.5 gm/dL
	Fasting lipids	Increased levels of TC, LDL, VLDL, and triglycerides, and normal or decreased HDL resulting in a high TC/HDL ratio
	Complement C3, C4, ANA ^a	Normal
	BSAg	Negative
	HCAb	
	Varicella titer	Those with negative titer need appropriate management with varicella infection and immunization.
Selected cases		
Blood test	ASOT CH50, Anti-dsDNA, ANCA, circulating immune complexes, Anti PLA2R HIV	Depending on clinical presentation, these tests may be needed to rule out the possibility of secondary NS; anti-PLA2R antibodies if membranous nephropathy suspected in an older child or adolescent
Skin test	Tuberculin test	To rule out tuberculosis before starting immunosuppression
Research purposes only	Urine biomarkers	Urine biomarkers that have shown promising results in differentiating FSGS from MCD include • uVDBP, • NGAL • A1BG, • CD-80 • A panel of 10 biomarkers [alpha-1 acid glycoprotein, <i>(continued on next page)</i>

Table 4
(continued)

Routine Labs	Test	Interpretation
		alpha-1 acid glycoprotein 2, alpha-1 microglobulin, alpha-1-B glycoprotein, fetuin-A, hemopexin, NGAL, prealbumin (transthyretin), thyroxine-binding globulin, and VDBP].
		• Additional urine biomarkers that appear promising include transferrin, UBA52, calretinin, CD 14, C9, A1AT and a few more ⁵⁸

Abbreviations: A1BG, α 1B glycoprotein; BSAG, hepatitis B surface antigen; HCAB, hepatitis C antibody; HDL, high-density lipoprotein; LDL, low-density lipoprotein; NGAL, neutrophil gelatinase-associated lipocalin; TC, total cholesterol; uVDBP, urine vitamin D binding protein; VLDL, very low-density lipoprotein.

^a Some centers do these selectively.

MANAGEMENT

Initial Management

The initial management of INS includes corticosteroid therapy, supportive care, parent education, and the treatment of complications, if present. Hospitalization should be considered in patients with anasarca, abdominal pain, fever, or for socio-economic considerations. About 5% of children undergo spontaneous remission within a week or two before treatment is initiated.²⁴ Spontaneous remission is also known to occur with the onset of measles or influenza B virus infection.²⁵ Such patients need close monitoring because a relapse is common and there is no evidence that their clinical course, including response to treatment, is any different as compared with the other children with INS.

Table 5
Indications and benefits of genetic testing in nephrotic syndrome

Indications	• Congenital (<3 mo old) or infantile (<1 y) NS • SRNS, especially in patients younger than 2 y • History of consanguinity • Nephrotic syndrome associated with syndromic features • Positive family history of SRNS/FSGS
Benefits	• Sparing of immunosuppression • Avoidance of a kidney biopsy • Prenatal counseling in future pregnancies • Renal transplant planning for post-transplantation recurrence • Discovering rare monogenic causes of SRNS that are amenable to cure of coenzyme Q10 leading to deficiency, occurring in 1% of cases of familial SRN;²³ coenzyme Q10 supplementation in these patients is effective for the treatment and cure of mitochondrial podocytopathies. ◦ Cubulin and <i>MMACH</i> genes may present with SRNS that responds to vitamin B12 supplementation.⁵⁹

Abbreviations: FSGS, Focal segmental glomerulosclerosis; SRNS, Steroid resistant nephrotic syndrome.

Corticosteroid therapy

Prednisone or its active metabolite prednisolone is the effective first-line therapy for INS. The first treatment regimen for corticosteroid administration for INS was established in 1979 by the ISKDC, which formed the basis for subsequent management protocols, including the current one. The ISKDC protocol consisted of prednisone 60 mg/m² per day for 4 weeks and 40 mg/m² 3 days a week for another 4 weeks.²⁶ After the ISKDC study, many studies have been published in the quest for optimization of therapeutic response to corticosteroids with minimal side effects.

Some studies have shown that the administration of corticosteroids for longer than 2 to 3 months lowers the risk of relapse²⁷ whereas others have reported no significant benefit of increasing the duration of therapy beyond 2 to 3 months.^{28–31} Based on recent data, the Cochrane Review concluded that the optimal duration of initial corticosteroid therapy should be 2 to 3 months and extending it for longer than 12 weeks does not significantly improve the outcome.³² The most recent Kidney Disease: Improving Global outcomes (KDIGO) guidelines recommend a total of 8 or 12 weeks of prednisone or prednisolone for children diagnosed with INS.³³

- Prednisone/prednisolone single dose at 60 mg/m²/d or 2 mg/kg/d (maximum of 60 mg/d) for 4 weeks followed by alternate-day single daily dose 40 mg/m² or 1.5 mg/kg (maximum 50 mg on alternate days) for another 4 weeks.

Or

- Prednisone/prednisolone single dose at 60 mg/m²/d or 2 mg/kg/d (maximum of 60 mg/d) for 6 weeks followed by alternate-day single daily dose at 40 mg/m² or 1.5 mg/kg (maximum 50 mg on alternate days) for another 6 weeks.

Four-week daily initial steroid therapy is not a drop-dead deadline for changing it to alternate days or defining steroid resistance. About 10% of patients with INS are known to achieve remission during the second month of daily steroid therapy.¹⁸ In a patient who is in partial remission at the end of 4-week treatment, the daily treatment may be extended up to a total of 6 weeks and once in remission, the alternate-day regimen is the same as for those responding within 4 weeks. Administration of a high-dose oral methylprednisolone³⁴ or coadministration of a second immunosuppressant such as cyclosporin A³⁵ or anti-inflammatory agent such as azithromycin³⁶ is not helpful. Intravenous (IV) methylprednisolone in the beginning of steroid therapy is also not very helpful.³⁷ However, in a study using the French protocol in children with no response after 4 weeks of daily prednisone therapy, the administration of 3 intravenous pulses of methylprednisolone (1 g/1.73 m²) induced remission in 10 of 22 (45%) patients.³⁸

In a study by Saadeh and colleagues, underdosing and risk of frequent relapses was reported when the prescribed dose of corticosteroid was based on body weight as compared with the body surface area.³⁹ However later studies have not reported any significant difference in clinical outcome between the two regimens.^{40,41}

Many country-specific practice guidelines about initial corticosteroid treatment in INS have been published. However, as shown in [Table 6](#) most of them are remarkably similar with clinically insignificant differences.

Diet

Protein: A relatively high protein intake of high biologic value is recommended, usually 130% to 140% of the normal daily allowance.

Sodium: All patients with NS have a positive sodium balance. This is generally attributed to a decrease in intravascular volume due to low plasma oncotic pressure,

Table 6

Initial corticosteroid therapy according to the various guidelines

Steroid Therapy	ISKDC ²⁶ 1979	Canadian ⁶⁰ 2014	Dutch ^{61,62} 2010	French ⁶³ 2017	German ⁶⁴ 2021	Indian ⁶⁵	Italian ⁶⁶ 2017	KDIGO 2021
Total duration (weeks)	12–24	12–24	12	18	12	12	12	8 or 12
<i>Dose and duration daily corticosteroid (weeks)</i>								
• 60 mg/m ² or 2 mg/kg daily	4	4–6	6	4	6	6	6	4 or 6
• 60 mg/m ² /EOD	-	-	-	8	-	-	-	-
• 45 mg/m ² EOD				2				
• 40 mg/m ² /EOD (1.5 mg/kg) OR 40 mg/m ² /3 d per week (1.5 mg/kg)	4	8–20 -	6 -	- -	6 -	6 -	6 (single dose) -	4 or 6 -
• 30 mg/m ²				2				
• 15 mg/m ²				2				
Maximum dose (mg) while on daily/EOD corticosteroid	80/60	60/40	80/?	60/?	80/60	60/40	60/40	
Single or two divided daily doses	Divided	Single	Single		Single	Single or divided	Single or divided for daily/single for EOD	
Tapering	No	Yes	No	yes	No	No	No	

Abbreviation: EOD, every other day.

causing activation of the renin–angiotensin system and increased renal sodium absorption, as well as the activation of ENaC and Na-K ATPase in cases with intravascular expansion. A “no added salt” diet generally works in most cases, though a rigid sodium restriction to approximately 1 mEq/kg per day is recommended in severe cases.

Fluids: Fluid restriction is not easy particularly in ambulatory settings, and it may not be necessary unless there is moderate or severe hyponatremia. INS patients with severe edema may have an increased osmolal gap secondary to non-Na⁺ and non-K⁺ osmoles due to plasma volume contraction. Severe fluid restriction in such patients could increase their risk of prerenal failure.⁴²

Diuretics

In ambulatory settings and depending on the severity of edema, a low-dose oral loop diuretic such as furosemide may be used alone or in combination with a distal tubular acting diuretic, such as amiloride or aldactone. In hospitalized patients, IV furosemide may be used with or without albumin, depending on the intravascular volume status. Patients with FeNa of $\geq 0.2\%$ may receive a diuretic alone whereas it is safer to coadminister a diuretic with IV salt-poor albumin in those with FeNa of less than 0.2% .⁴³ Diuretics should be discontinued promptly if the patient develops an intercurrent illness associated with fever, poor oral intake, vomiting or diarrhea because of an increased risk of thrombosis and acute kidney injury.

Albumin

IV albumin is reserved for patients with severe edema. Albumin infusion followed by IV furosemide is usually effective in refractory cases, but repeated infusions may be needed as the infused albumin is rapidly excreted in the urine. Patients, particularly those who may have hypervolemia, should be watched for complications such as hypertension, pulmonary edema, or congestive cardiac failure.

Infection risk assessment

Children with NS are at increased risk of pneumonia, peritonitis, or sepsis due to reduced cellular and humoral immunity, and the administration of immunosuppressant medications. *Streptococcus pneumoniae*, which used to be one of the commonest infections is uncommon nowadays because of the polyvalent pneumococcal vaccination.

The assessment of Varicella Zoster immune status should be included in the initial workup. Those susceptible need Zoster immunoglobulin, sometimes with acyclovir, depending on severity of the infection. No live vaccine should be administered during daily, high-dose steroid therapy or administration of alkylating agents. According to the American Academy of Pediatrics Red Book, 2021, the recommendations on live vaccines are as follows:⁴⁴

- Children receiving ≥ 2 mg/kg/d of prednisone or its equivalent, or ≥ 20 mg/d (daily) if they weigh 10 kg or more for 14 days or more should not receive live-virus vaccines until 4 weeks after the discontinuation of treatment.
- Children receiving less than 2 mg/kg/d of prednisone (*daily or alternate days*) or its equivalent or less than 20 mg/d if they weigh 10 kg or more can receive live-virus vaccine during steroid treatment.

Killed vaccines such as pneumococcal conjugate (as well as 23-valent pneumococcal polysaccharide vaccine) can be given anytime. Live poliomyelitis vaccine should be substituted by inactivated vaccine and the live vaccine should not be given

to siblings or other patient contacts during daily steroid treatment. Patients can have annual influenza vaccination.

Activity

We encourage physical activity as tolerated. This helps in mobilizing edema fluid and reduce the probability of venous thrombosis, muscle weakness, and osteoporosis.

Parent education

Parent education about daily urine dipstick testing at home helps monitoring for remission or relapse and thereby allows a prompt intervention and adjustments in corticosteroid administration. Education on dietary sodium restriction goes a long way in alleviating edema and may set up dietary lifestyle changes, which is particularly helpful in those who end up with a prolonged clinical course and are at risk of recurrent edema and or hypertension. Similarly, counseling on the risk of dehydration with gastroenteritis or poor oral intake due to abdominal symptoms, or increased risk of infection can be lifesaving.

Treatment of Relapse(s)

Approximately 80% to 90% of patients with INS relapse after initial response and 40% to 50% of these become steroid dependent (SD) and have frequent relapses (SD/FR). The current KDIGO recommendation for the treatment of a relapse is as follows:³³ Prednisone/prednisolone single dose at 60 mg/m²/d or 2 mg/kg/d to a maximum of 60 mg/d until in remission for ≥ 3 days followed by 40 mg/m² or 1.5 mg/kg (maximum 50 mg on alternate days) for ≥ 4 weeks.

The two commonly used strategies for the management of children with FR/SD NS are given below.

Long-Term Low-Dose Corticosteroid

In FR/SD patients with no significant corticosteroid-related side effects, an alternate-day, low-dose corticosteroid for 6 months or longer may be used to sustain a remission.^{45,46} The recommended dose is ≤ 0.5 mg/kg on alternate days. In a recent study, a daily administration of a low dose (0.2–0.3 mg/kg) over a 12-month period was reported to be more effective as compared with alternate days (0.5–0.7 mg/kg alternate day).⁴⁷

Corticosteroid-Sparing Agents

In patients who need a high dose of corticosteroid to stay in remission or have steroid side effects, other medications that may be considered include calcineurin inhibitors (cyclosporin or Prograf), mycophenolate mofetil, levamisole, cyclophosphamide and rituximab. Further details about their use are listed in [Table 7](#).

Prevention of Relapse

Relapses in INS are often precipitated by URIs. In a study of 36 children with FRNS on alternate-day, low-dose maintenance corticosteroid, it was shown that giving the same dose daily for 5 days significantly reduced the risk of a relapse (95% CI: –4.03 to –2.57; 36 participants).⁴⁸ These findings were validated by more studies^{32,49,50} and this concept was later extended to children who are not on corticosteroids. Thus, in a study by Abeyagunawardena and colleagues⁵⁰ it was shown that a short course of daily corticosteroids (0.5 mg/kg/d) during a URI significantly reduces the frequency of URI-induced relapse in patients with steroid-responsive NS.⁵¹ KDIGO currently recommends that for children with FRSD NS, who are currently taking alternate-day, low-dose corticosteroid or are off steroids receive daily corticosteroid

Table 7
Steroid-sparing medications for steroid-dependent/frequent relapsing nephrotic syndrome

Medication		Comments
CNI	Cyclosporin or tacrolimus	• Used for steroid-resistant as well as frequent relapsing/steroid-dependent cases. Generally used as first-line therapy for the latter. • CNIs are nephrotoxic, need to be given for a prolonged period of time, and nephrotic syndrome tends to recur when medication is stopped. • A minimum of 6 mo of treatment is required to determine whether the patient is responsive to the CNI. • Optimum duration of therapy with CNIs among responders is debatable although a minimum of 12–24 mo is frequently cited.⁶⁷ • Kidney biopsy to assess CNI-induced nephrotoxicity may be needed in selected cases after prolonged administration.⁶⁸
	Mycophenolate mofetil	• Fewer side effects as compared to CNI, CYP, and tacrolimus. • Is being increasingly used in frequent relapsing and steroid-dependent cases. • Recommended also for children who attain full remission on CNI treatment for ≥ 1 y, thus allowing for discontinuation of CNI.⁶⁷
	Levamisole	• An anthelmintic agent with low toxicity profile; it is not available in the USA.
	CYP	• Generally avoided because of its toxicity but can be used effectively if other medications are not available. • Monitoring for side effects during administration is essential.
Monoclonal antibodies	Rituximab	• Chimeric anti-CD20 antibody that causes depletion of B cells. • Relatively new medication with potentially serious side effects and should be used cautiously and selectively. • May be considered in steroid-resistant as well as frequent relapsing/steroid-dependent cases. • Recent study by Basu and colleagues that included 176 pediatric patients with steroid-dependent nephrotic syndrome reported a higher 12-mo

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Table 7
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Medication	Comments
	relapse-free survival rate in the rituximab group than in the tacrolimus group (90.0% vs 63.3%) as well as a lower 12-mo cumulative corticosteroid dose in the rituximab group. ⁶⁹
Ofatumumab abatacept, adalimumab fresolimumab	• Newer anti-CD20 monoclonal antibodies with little or no data on their use in nephrotic syndrome

Abbreviations: CNI, calcineurin inhibitor; CYP, cyclophosphamide; USA, United States of America.

0.5 mg/kg/d during URI or any other infections for 5 to 7 days to reduce the risk of a relapse.³³

Other Supportive Care

Some patients with INS, particularly those with FR/SD clinical course, may need clinical intervention for the following:

Behavioral Issues

The occurrence of behavioral issues with steroid administration is not uncommon and these may include anxiety and depressive or aggressive behavior. It is worse in those with abnormal behavior at baseline. According to a study by Soliday and colleagues, the behavioral changes occur mostly at prednisone doses of 1 mg/kg every 48 hours or more.⁵² Depending on the severity, an adjustment in steroid dosage, administration of a steroid sparing agent, or an appropriate referral should be considered in such patients.

Hypertension

Hypertension is rare at presentation in INS, particularly in those who respond to steroid therapy. A few patients may become hypertensive during steroid therapy, and the risk of hypertension increases in those who have recurrent relapses or become steroid resistant. Administration of CNI and the onset of CKD further increases the risk of hypertension in such cases. Angiotensin converting enzyme inhibitors or angiotensin II receptor blockers are preferred because of their additive antiproteinuric benefit and ability to slow progression of renal impairment. Other options include calcium channel blockers or beta-blockers.

Hypercoagulability

The risk factors for vascular thrombosis in NS include hemoconcentration, thrombocytosis, hypercoagulability, dyslipidemia, and hypovolemia. Low-dose aspirin or dipyridamole, or the administration of an anticoagulant may be needed in high-risk patients, particularly those with an episode of a thromboembolic complication. Prompt treatment of sepsis, prevention of intravascular volume depletion and hemoconcentration, regular ambulation, and avoidance of central catheters help reduce the risk of thromboembolism.

Hyperlipidemia

Hyperlipidemia is a common occurrence in INS and in the past its presence was included in the definition of NS. It is mainly a result of an increased hepatic synthesis

of cholesterol, triglycerides, and lipoproteins and a decreased catabolism of lipoproteins caused by decreased lipoprotein lipase. Patients can have increased levels of total cholesterol, low-density lipoprotein, very low-density lipoprotein, triglycerides, and normal or decreased high-density lipoprotein resulting in a high TC/HDL ratio. These resolve soon after the resolution of proteinuria and as such there is no need for routine administration of statins in INS but should be considered in those with refractory NS.⁵³

NEPHROTIC SYNDROME IN ADULTS

Adults may have a new-onset INS or continue having relapses from the disease diagnosed during childhood. Recent literature shows that a more severe form of SS INS in childhood may continue into adult life. In a study that included 102 adults with NS in childhood, 43 (42.2%) experienced at least one relapse of NS in adulthood.⁵⁴ In another recent study, in 39 adult patients with a mean age of 22.8 (range 18.0–30.9) years with childhood-onset steroid sensitive nephrotic syndrome (SSNS) at a mean age of about 8 years (6–11.1) years, 12 (31%) had active disease in adulthood. All had SD/FR NS in childhood.⁵⁵ As such, proper counseling at the time of discharge from the pediatric clinic is recommended.

SUMMARY

NS is diagnosed by the presence of significant proteinuria in an edematous child. A good clinical assessment with a few laboratory tests helps rule out secondary causes for NS. All patients with INS should be treated with 2 to 3 months of corticosteroid therapy. Most patients go into remission but have relapse(s) that may need additional immunosuppression, including administration of steroid-sparing agents. The risk of a relapse can be decreased by a short 5-day course of steroid therapy during URI. Complications associated with the disease or medication side effects are not uncommon. Some patients continue to have relapses in adult life.

CLINICAL CARE POINTS

- Intravenous albumin and furosemide should be used only in patients with severe edema not amenable to salt and water restriction or oral diuretic administration.
- The duration of initial corticosteroid therapy for more than 12 weeks is not necessary.
- No live vaccine should be administered during high-dose steroid therapy.
- An alternate-day, low-dose (≤ 0.5 mg/kg) steroid for 6 months or longer can be used to sustain a remission.
- A short 5- to 7-day course of daily steroids at the onset of URI lowers the risk of a relapse.
- The commonest indication for kidney biopsy is the lack of response to initial corticosteroid therapy.
- Treatment with a non-steroidal immunosuppressant should be considered in patients with frequent relapses/steroid dependence or steroid toxicity.

DISCLOSURE

The authors have nothing to disclose.

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